

KELT-20b

R-2390

Benchmark run · workflow W8-benchmark-deep · record deep - bench_KELT-20_260716 · created 2026-08-20T23:35:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461237.7744 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.04
Transit depth (Rp/Rs)^2	0.75 +/- 0.7 [%]
Orbital Inclination (inc)	81.57 +/- 2.64
Airmass coefficient 1 (a1)	23.38 +/- 2.76
Airmass coefficient 2 (a2)	-3.0 +/- 0.82
Scatter in the residuals of the lightcurve fit is	12.77 %
Best Comparison Star	#2 - [171, 465]
Optimal Aperture	2.9
Optimal Annulus	13.12
Transit Duration (day)	0.03 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.087 $\pm$ 0.04 vs 0.1144 (deviation 0.46 σ)
Duration fit vs published	0.03 d vs 0.1471 d (deviation 2.43 σ)
Mid-time O-C	-5.5 min
Depth SNR	1.4
Residual scatter	12.77 %

Light curve

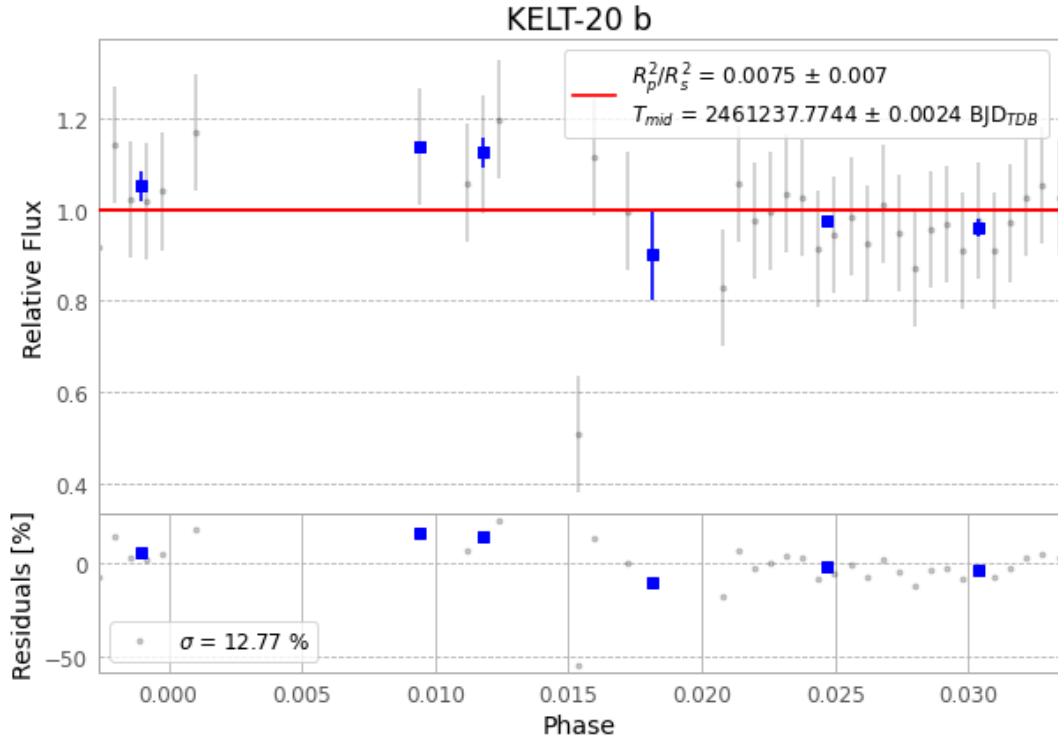


Figure 1 — FinalLightCurve_KELT-20 b_2026-07-15.png (SHA-256 0b2540d51b1742ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [249, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b651ba3ce03dc11c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), KELT-20260716061515.FITS → KELT-20260716091516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-20-260716.log (8d04019a71f1...), FinalLightCurve_KELT-20 b_2026-07-15.png (0b2540d51b17...), FinalLightCurve_KELT-20 b_2026-07-15.csv (4f617d41329f...)

Compute resources

Wall time 195.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:35Z.